

Ceph Deployment Guide: Step-by-Step with `cephadm`

This guide provides a straightforward path to deploying a basic Ceph cluster using `cephadm`, the modern and recommended orchestration tool for Ceph.

📄 1. Prerequisites

Before you begin, ensure all nodes in your cluster meet the following requirements:

- * ****OS:**** A modern Linux distribution (e.g., Ubuntu 20.04+, CentOS 8+, Debian 10+).
- * ****Python:**** Python 3 must be installed.
- * ****Container Engine:**** Podman (recommended) or Docker.
- * ****Time Sync:**** Chrony or NTP must be active for cluster consistency.
- * ****LVM2:**** Required for OSD provisioning.
- * ****Network:**** Static IP addresses and SSH access between nodes.

🚀 2. Install `cephadm`

On your first host (the bootstrap node), install the `cephadm` utility:

```
``bash
# Download the cephadm script
curl --silent --remote-name --location
https://github.com/ceph/ceph/raw/quincy/src/cephadm/cephadm

# Make it executable
chmod +x cephadm

# (Optional) Install cephadm to your path
sudo ./cephadm add-repo --release reef
sudo ./cephadm install
``
```

🏗️ 3. Bootstrap the Cluster

Run the bootstrap command on your first node. This creates the initial Monitor and Manager daemons.

```
``bash    inet 172.21.204.100  netmask 255.255.240.0  broadcast
172.21.207.255
sudo cephadm bootstrap --mon-ip 172.21.204.100
``

* **Output:** This will provide you with the **Dashboard URL**,
**Username**, and **Password**. Keep these safe!
* **Result:** You now have a "cluster of one."
```

🐙 4. Enable the Ceph CLI

To run `ceph` commands directly from your shell:

```
```bash
sudo cephadm shell
OR install the ceph-common package
sudo ./cephadm add-repo --release quincy
sudo ./cephadm install ceph-common

Verify health
ceph health
```
```

+ 5. Add More Hosts

To make the cluster resilient, add more nodes. First, copy the SSH public key to the new nodes:

```
```bash
ssh-copy-id -f -i /etc/ceph/ceph.pub root@<NEW_HOST_IP>
```
```

Then, add the host to the cluster:

```
```bash
ceph orch host add <NEW_HOSTNAME> <NEW_HOST_IP>
```
```

🗄️ 6. Add Storage (OSDs)

Ceph can automatically discover and use available raw drives.

```
```bash
See available devices
ceph orch device ls

Tell Ceph to use all available raw devices (be careful!)
ceph orch apply osd --all-available-devices
```
```

🖥️ 7. Access the Dashboard

Open the URL provided during bootstrap in your browser. You can monitor cluster health, OSD status, and manage pools through this web interface.

🗒️ Summary Checklist

1. [] Prerequisites met on all nodes.
2. [] `cephadm` installed on bootstrap node.
3. [] `cephadm bootstrap` completed.

4. [] SSH keys distributed to other nodes.
5. [] Additional hosts added via `ceph orch`.
6. [] OSDs deployed on available drives.